

Grouping-rule evidence: capacity signals for the sectoral rigidity set

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Evidence annexe to the grouping-rule decision point of `docs/WORKPLAN_SENSITIVE_PRICES.md` (v7) and the ADR-0022 open item (“a rule is preferred to a hand-picked list, to forestall the tuning objection”). The executed `matrix_5x3_v9` two-group cells carry two *rules* (the seven programme sectors; the largest half by baseline employment) whose choice flips the sign of the employment effect. This note collects the external, pre-run evidence that a *principled* third rule can be stated from, and states that rule. Nothing here has been run; the rule is for ratification and then preregistration with the design.

1 The rule (candidate, for ratification)

A sector’s labour supply is rigid ($\text{eta}_{s,i} = 0$) iff (i) its programme incidence share is at least 1 % of the programme total AND (ii) its skilled-worker-shortage signal reaches the goods-producing average or above.

Rationale, in one sentence: the rigidity of a sectoral labour supply for the programme’s purposes is the *elasticity* of its labour market, and the best available external proxies for short-run inelasticity are persistent vacancies and reported skills shortages — not the level of capacity utilisation, which is cyclical and was falling in 2024.

The rule is computable before any run: (i) is read off the 2024 impulse data, (ii) off the 2024 shortage reports below. Its output — the rigid set — is a model *input*, not an artefact of the results.

2 Programme incidence (i) — from the data

From `cbase2/data_raw/impulses.csv`, 2024 row (71 VGR sectors, $G0 = 40,300$ EUR m): the programme’s incidence is concentrated in seven sectors.

Sector	2024 impulse share
Specialised construction works	63.4 %
Rubber and plastics products	9.4 %
Ceramic products, processed stone and clay	8.0 %
Chemicals and chemical products	7.6 %
Glass and glassware	6.1 %
Machinery	4.3 %
Electrical equipment	1.3 %

Condition (i) selects exactly these seven (the next sector in the ranking has zero incidence).

3 Capacity signals (ii) — 2024 evidence

Indicator	Value (2024)	Coverage	Source
Skilled-worker shortage, Bauhauptgewerbe (Q4/2024 survey, October)	28.9 % of firms report production restricted by skills shortage (27.7 % in July; peak 37.0 % Q4/2022)	WZ broadly building construction + civil engineering + demolition + specialised works (Destatis Bauhauptgewerbe grouping)	KfW-ifo-Fachkräftebarometer December 2024 (KfW Research / ifo Institut), p.2
Skilled-worker shortage, Verarbeitendes Gewerbe aggregate (Q4/2024)	20.6 % (peak 44.5 % Q3/2022; long-run mean 9.7 %)	manufacturing broad	KfW-ifo-Fachkräftebarometer, December 2024
Skilled-worker shortage, Gummi- und Kunststoffwaren (Q4/2024)	33.1 %	one of the six material sectors (rubber/plastics)	KfW-ifo-Fachkräftebarometer, December 2024
Skilled-worker shortage, Metallerzeugung/-bearbeitung (Q4/2024)	10.3 %	comparator: metal processing not programme-relevant	KfW-ifo-Fachkräftebarometer, December 2024
Open positions, Baugewerbe (Q4/2024)	108,000 (national: 1.40 m; Q4/2022 peak ~2.0 m)	construction sector	IAB-Stellenerhebung, IAB-Monitor Arbeitskräftebedarf 4/2024 (IAB-Forum, 14 Mar 2025)
National Vakanzrate (Q4/2024)	3.2 % (immediately open positions per 100 demanded)	all sectors; 2.9 % in Q3/2024	IAB-Monitor Arbeitskräftebedarf 4/2024
Order backlog, Bauhauptgewerbe (June 2024)	3.7 months overall (Hochbau 3.5, Tiefbau 4.1, sonstiger Tiefbau 4.5)	ifo Konjunkturumfragen	ifo / bauletter 1 Jul 2024; declining from 4.8 months (Jan 2022)
Working-time accounts, Baugewerbe (2023 -> 2024)	20 % -> 62 % of firms (Hesse), the largest rise of any sector	IAB-Betriebspanel 2024	IAB-Betriebspanel Report Hessen 2024 (HNA/iwak)
Personnel demand persistence, Baugewerbe (H1 2024)	54 % of firms with hiring need (3rd highest); of those, 72 % left positions unfilled	IAB-Betriebspanel, Brandenburg 2024	Betriebspanel Brandenburg 2024 (IAB)

4 What the indicators mean, in plain words

The four sources are three national series and one pair of regional panels. None of them measures a labour-supply elasticity; together they triangulate *how hard it is to get workers quickly* in the programme’s sectors, which is the short-run margin the rigidity assumption ($\eta_{s,i} = 0$) is about.

- **IAB-Stellenerhebung** (Institute for Employment Research, quarterly establishment survey): firms report all positions they are trying to fill, including the ones not registered at the employment agency. Two quantities matter: the *stock* of open positions (Baugewerbe 108,000 in Q4/2024, of 1.40 million nationally) and the **Vakanzrate** — immediately open positions per 100 workers demanded (employed plus open), the standard national tightness measure (3.2 % in Q4/2024). Reading: even after a year in which vacancies fell 19 % (the 2024 downturn), construction still holds a top-five stock of positions that firms keep failing to fill — a *supply* symptom, not a demand symptom.
- **KfW-ifo-Fachkräftebarometer** (KfW Research / ifo Institut, quarterly survey of about 9,000 firms across manufacturing, the building main trades, trade and services): one question, “is your business currently restricted by a lack of skilled workers?”; the number is the **share of firms** answering yes. Q4/2024: 28.9 % in the Bauhauptgewerbe against 20.6 % in manufacturing broadly (plastics — rubber/plastics, one of the six material sectors — 33.1 %, the most affected manufacturing branch in the report). Construction runs at roughly 1.4x the manufacturing rate at a moment when manufacturing itself is slack. Note the measure is firms’ self-reported current restriction and needs a cyclical caveat (it fell from a 37.0 % peak in Q4/2022).
- **ifo Konjunkturumfragen — Auftragsbestand** (order backlog in months): firms state how many months of work their order books cover at current capacity. 3.7 months (June 2024) means: were new orders to stop, construction firms still have 3.7 months of scheduled work. This is the demand-side/capacity-cushion indicator, and it fell from 4.8 months (early 2022). It is in the table deliberately, as the *counter-signal*: it documents that the 2024 story is not “order books bursting”, so the rigidity claim must rest on the labour-elasticity signals (the other three), not on utilisation.
- **IAB-Betriebspanel** (annual establishment survey): two regional 2024 state reports (Hesse, Brandenburg) are used as directional evidence, not national estimates. (i) Working-time accounts in the Baugewerbe jumped from 20 % to 62

% of firms 2023->2024 in Hesse (37 % -> 58 % West): when order books are volatile, firms keep headcount stable and buffer through *hours* — hours are the flexible margin, employment the rigid one, which is precisely the vertical-supply assumption. (ii) In Brandenburg, 54 % of construction firms had a hiring need in H1/2024 (third-highest of all sectors) and 72 % of those left positions unfilled — recruitment failures persist even in a weak year.

5 What the rule does with them

The model’s rigidity is a property of the *slope* of a sector’s labour supply (its elasticity), never measured by any of these series. The rule therefore only uses them to *select the set*: a sector is rigid when it concentrates the programme (incidence ≥ 1 %) **and** sits at or above the goods-producing average on the skills-shortage signal. The value of **eta_s** itself comes from the supply arm (ADR-0023), not from this table. What the table establishes, before any model run, is that the seven programme sectors — not some hand-picked list — are the sectors whose labour markets show the shortage pattern the rigidity assumption formalises.

6 What the table says, and its limits

- **The capacity-pressure reading survives the 2024 downturn — with a caveat.** Order backlogs fell from the 2022 peak (4.8 -> 3.7 months) and open positions fell 19 % y/y, i.e. the 2024 demand side was weak. But the signals the rule uses are the *elasticity* signals: skills-shortage reports and vacancy persistence. Construction still reports 28.9 % (8.3 pp above manufacturing), plastics 33.1 %, and 108,000 open construction positions; working-time accounts doubled in a year — firms buffer volatile order books through hours, which is consistent with headcount that cannot expand at will.
- **Provisional output of the rule.** With the collected signals, condition (ii) is firmly met by specialised construction and rubber/plastics (both above the manufacturing average), plausibly met by ceramics, chemicals and glass, and *unverified* for machinery and electrical equipment (2024 sub-branch shortage shares not retrieved). A defensible pre-run specification registers **two** evidence-rule variants: the five-sector core {specialised construction, plastics, ceramics, chemicals, glass} (94.5 % of programme incidence) and the full seven under a weaker threshold (≥ 20 %, i.e. the manufacturing average). The five-sector set is the default; the seven-sector set tests the threshold’s sensitivity.
- **Mapping caveat.** The model’s sectors are VGR (Destatis 2019 IO classification); the reports above use WZ/NACE groupings. “Specialised construction works” ~ WZ 43, and the Destatis Bauhauptgewerbe indicator used by ifo/KfW covers WZ 41.2, 42.x, 43.1, 43.9 — a superset. The material suppliers map to their WZ classes up to aggregation. This is approximation at the 2-digit level, recorded rather than hidden.
- **Vintage.** All signals are 2024, matching the programme vintage (impulses.csv 2024 row, G0 = 40,300 EUR m at 2019 prices). The shortage shares are firm-level survey shares (KfW-ifo), not labour-market elasticities; they are the best public proxies and are used only to *select* the rigid set, never as estimates of **eta_s** itself.
- **Not a claim about wage policy.** As in ADR-0022: a welfare difference across rules is not a statement about who should earn what; the rule only answers which sectors’ supply curves are treated as inelastic.

7 Next steps

1. Ratification of the rule wording (and the five-vs-seven threshold).
2. The grouping-rule batch design (rules: uniform / programme / evidence-5 / evidence-7 / largest-half, x F1/F3 at the settled **eta_s** level), rows **planned** in **registry/scenarios.csv**, preregistered (ADR-0006) citing this note as the rule’s evidence, executed, and the employment sign compared across rules — the preregistered signature is that the *rule*, not the elasticity level, is the sign-relevant choice.
3. The same evidence ranges become the probability measure of the sectoral Sobol (IDEA-0003).

8 Revision Log

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